

Lecture 7

Compactness

- $A \subset M$, A is compact iff $\forall (a_n)$ in A , $\exists$ a subsequence (a_{n_k}) that converges to a point in A
- $[a, b]$ is compact
- Every closed subset of a compact is compact
 - Suppose $K \subset A$ and A is compact. Any sequence in K is also a sequence in A , so there exists a convergent subsequence in K . K also must contain this limit as K is closed, so K is compact
- Every compact is closed and bounded
 - Proof by contradiction.
 - Suppose not. Then, A is compact and it's not closed or not bounded.
 - If A is not bounded, $p \in A$, look at successively larger $M_r(p)$.
 - Not bounded means $\forall r$, exists $a_n \in A \setminus M_r(p)$
 - Convergent subsequence, but finite distance from p
 - The above kind of doesn't make sense
 - If A is not closed, some sequence converges to something not in A .
 - Convergent subsequence also doesn't converge to something in A , contradicting the definition of compactness
 - Converse is false
 - $\{x \in \mathbb{Q} : \pi < x < 2\pi\}$
 - Closed as subset of $\mathbb{Q}$, also bounded as all values are less than 2π (not saying it's bounded by 2π though)
 - Is not compact, as limits of some sequences are not contained in $\mathbb{Q}$

Heine Borel Theorem: $A \subset \mathbb{R}^m$ is compact iff closed and bounded in $\mathbb{R}^m$

In $\mathbb{R}$

- $A \subset \mathbb{R}$ is compact iff closed & bounded in $\mathbb{R}$
- $\Rightarrow$ already proven
- $\leq$ find a compact set in $\mathbb{R}$ that contains A
 - $\exists [a, b]$ s.t. $A \subset [a, b]$
 - Closed and bounded subset of compact is compact, so A is compact
- Holds in $\mathbb{R}$, not necessarily any complete metric space
 - Question on midterm: a complete, closed, bounded set is not necessarily compact

Defining metrics

- M_1, M_2 have metrics d_1, d_2
- 3 choices for metrics
 - $d_E((x_1, x_2), (y_1, y_2)) = \sqrt{d_1(x_1, x_2)^2 + d_2(y_1, y_2)^2}$
 - $d_{\max}((x_1, x_2), (y_1, y_2)) = \max\{d_1(x_1, x_2), d_2(y_1, y_2)\}$
 - $d_{\text{sum}}((x_1, x_2), (y_1, y_2)) = d_1(x_1, x_2) + d_2(y_1, y_2)$
- $d_{\max} \leq d_E \leq d_{\text{sum}} \leq 2d_{\max}$
- Convergence w.r.t to one means convergence w.r.t to others

If $A \subset M_1, B \subset M_2$ and both are compact, then $A \times B$ is compact w.r.t. to all 3 metrics

- We have some sequence (a_n, b_n) in $A \times B$
- $(a_n) \subset A$ has convergence subsequence, can be denoted (a_{n_k})
 - $(a_{n_k}) \rightarrow a$ as $k \rightarrow \infty$

- However, we cannot guarantee (b_{n_k}) converges. Hence, we take a convergent subsequence $(b_{n_{k_l}})$.
 - $(b_{n_{k_l}}) \rightarrow b$ as $l \rightarrow \infty$
- Since $(a_{n_k}) \rightarrow a$, $(a_{n_{k_l}}) \rightarrow a$
- Hence, (a_n, b_n) has the convergent subsequence $(a_{n_{k_l}}, b_{n_{k_l}})$, which converges to (a, b)
- Thus, $A \times B$ is compact
- $[a_1, b_1] \times [a_2, b_2] \times \dots \times [a_n, b_n] \in \mathbb{R}^m$ is compact
 - Closed, bounded in $\mathbb{R}^m$
 - Every set is contained in some box, which is compact

If M_1 and M_2 are compact, then $M_1 \times M_2$ is compact

- Take sequence in M_1
- Follow by some point

Uniform Continuity

- $f : M \rightarrow N$ is uniformly continuous iff $\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in M$ we have $d_M(x, y) < \delta \Rightarrow d_N(fx, fy) < \varepsilon$
- Contrast with normal continuity, which states $\forall \varepsilon > 0 \forall x \in M \exists \delta$ vs $\forall \varepsilon > 0 \exists \delta \dots$ for uniform continuity
- Same δ for all x 's, vs. δ can depends on x
- Example of continuous but not uniformly continuous is $y = x^2$
 - Ratio between ε and δ will not be bounded
 - If you want a bounded function, think about $\sin(x^2)$, which has waves that are increasingly compressed, can see how $\Delta y \gg \Delta x$

Theorem 2: If M is compact, then uniform continuity = continuity

- Suppose $f : M \rightarrow N$ is continuous and M is compact
- Suppose f is not uniformly continuous
- $\exists \varepsilon > 0$ s.t. no δ works at all $x \in M$ & f is continuous
 - i.e. $\forall \frac{1}{n} \exists p_n, q_n \in M$ s.t. $d_M(p_n, q_n) < \frac{1}{n}$ but $d_N(fp_n, fq_n) \geq \varepsilon$
- $\exists p_{n_k} \rightarrow p \in M$
- q_n is forced to converge to p since $d_m < \frac{1}{n}$
- Contradiction to ordinary continuity, so f is uniformly continuous

Theorem 3: If $f : M \rightarrow N$ is a continuous bijection and M is compact, then f is a homeomorphism

- Suppose not. (WE love proof by contradiction)
- There exists a bijection f^{-1} , so we suppose f^{-1} not continuous
- $\exists \varepsilon > 0$ and $\forall \frac{1}{n} \exists x_n, y_n \in N$ s.t. $d_N(x_n, y_n) < \frac{1}{n}$ but $d(f^{-1}(x_n), f^{-1}(y_n)) > \varepsilon$
- WLOG, assume x_n, y_n converge in N , $x_n \rightarrow x$, $y_n \rightarrow y$ so $x = y$ as $d_N(x_n, y_n) \rightarrow 0$
- For subsequence $f^{-1}(x_n) \rightarrow a$, $f^{-1}(y_n) \rightarrow b$, $a \neq b$
- f^{-1} is not a function, which is a contradiction

Compactness, Circles, and Intervals

- A circle and a half open, half closed interval cannot be homeomorphic because one is compact, whereas the other isn't
- If 2 points are removed, the circle turns into 2 disconnected segments, whereas the interval turns into 3.